MD

Again, it comes down to identity. If in a previous art context the focal point for value creation was the individual, there's all this room to explore what new identity creation and formation can look like in this space. Holly refers to it as identity play, as opposed to intellectual property. That's the new opportunity. And it just feels perfectly calibrated for these tools.

BR

Right, but the tools are definitely not done. So here's a call to action to whoever ends up reading this: Help us out, build these tools with us.

MINDY SEU



The Metaverse is a Contested Territory

How the sci-fi of "Geek Gods" is creating the future we don't want.

Legend has it that six months before Facebook would rebrand as Meta, Mark Zuckerberg passed around a 2011 science-fiction novel to his employees. "This is it," he supposedly said. "This is what we're going to make."

The page-turner-turned-guidebook was *Ready Player One*, Ernest Cline's first book and a tale of outcast-to-hero, rags-to-riches, and geek-wins-the-girl. In the year 2045, the world is an overpopulated dystopia with food shortages and a never-ending recession. To escape the dread of "real life," people live, work, and entertain themselves in the OASIS, a virtual reality and ubiquitous internet created by the socially inept video game developer James Halliday, a "god among geeks, a nerd uber-deity."[1]

When Facebook began in 2004, it was billed as a platform for connectivity. Now, it anchors the oligopolies that shape our on- and offline experiences. Due in part to campaigns against privacy and tracking, Facebook rebranded as Meta to escape the negative connotations linked to the company. Its new mission was virtuality, claiming the "metaverse" as its latest territory. In fact, the term "metaverse" originated in the 1992 science-fiction novel *Snow Crash* as a portmanteau of "meta" and "universe." This root word—universe—is the territory at stake: a single, universal virtual world. As new technologies emerge, these *fictions*, as its etymological root suggests, *form* and *build* the scaffolding for how they might be applied.[2] New fictions, or highlighting old fictions less valorized, might reveal new virtues in these virtualities and contest the monolithic ownership of this new territory.[3]

In *Ready Player One*, OASIS is an open-source reality, a "holodeck for the home," where all global netizens have access to otherworldly landscapes, entertainment, and an escape.[4] Its biggest selling point? It's

1. Ernest Cline, *Ready Player One* (New York: Random House Publishing Group, 2011), 52.

2. The root of "fiction" comes from the Latin *fictionem* (a fashioning or feigning) and *fingere* (to shape, feign), originating in the Proto-Indo-European *dheigh* (to form, build).

3. The root of "virtual" is "virtue" or *virtu* (manliness) and *vir* (man).

4. Ernest Cline, *Ready Player One*, 59.

free. "Even though it was initially marketed as a new kind of massively multiplayer online game, the OASIS quickly evolved into a new way of life."[5] It comes under threat through Innovative Online Industries, otherwise known as IOI, who believe the ubiquitous online environment was never properly monetized and wish to implement widespread advertising while revoking user anonymity and free speech. Reading this, the irony is uncanny—even before their rebrand, Meta proclaimed itself to be a utopic, OASIS-like social platform where users can freely convene and connect. Beneath the surface, of course, its practice has been to privatize and homogenize our online spaces, commodify users, and sell their data. Meta's supposed commitment to freedom of speech has furthermore been tainted by shadowbans of those deemed overly political and overly sexualized. In this way, perhaps Zuckerberg's excitement over the novel was the ubiquity of OASIS—the global takeover—sold through the veil of utopia.

The metaverse's development is often linked to advancing virtual reality technology because to inhabit a metaverse, one must be connected in a virtual world. However, we should not equate Zuckerberg's Metaverse with virtual reality overall. To do so would conflate one company's simulation with the multiplicity of ongoing and past explorations.

The history of VR development began with researchers wanting to understand the physiology of human eyes. With the military driving the development of these tools, William E. Bradley's 1961 Headsight for Philco was the first head-mounted display (HMD), created for military use of video surveillance. Later that decade, Ivan Sutherland and his student Bob Sproull invented The Sword of Damocles, a militaristic tool funded by the Department of Defense and the CIA.[6]

By 1984, Jaron Lanier popularized the phrase "virtual reality" and his team at the VPL (Virtual Programming Languages) Research Lab invented the Eyephone, a HMD that featured head tracking. Virtual Reality fully entered the public consciousness and pop culture in the 1990s. Nintendo's 1995 Virtual Boy is one of the first consumer VR devices released, but it was discontinued a year later, in part due to its low-fidelity and the motion sickness it seemed to cause.[7] By 1999, Philip Rosedale founded Linden Lab, a developer of VR hardware like The Rig, a VR device users could wear on their shoulders. Like the aforementioned devices, development was difficult, but they went on to create virtual worlds, like the well-known *Second Life*.[8] Oculus Rift was announced in 2012 (first crowd-funded on Kickstarter before Facebook's acquisition), and Google Cardboard came out in 2014.[9]

Now, Meta continues to pour billions of dollars into the development of VR headsets, aiming for the ever-changing cutting edge of size, weight, and speed. In *Ready Player One*, players access the virtual environment through a wireless, one-size-fits-all OASIS visor that's smaller than a pair of sunglasses. With harmless, low-powered lasers the image is drawn onto its wearer's retinas, immersing their entire field of vision in the online world.

With the evolution of VR technology, we see the same story mirrored across sci-fi and reality: the hardware gets smaller and lighter, while the software allows for more pixels and more frames per second. However, it would be a mistake to conflate the development of VR software and hardware with virtual reality itself. In Rendering Worlds, a course co-taught at The New School by Deborah Levitt and Sarah Rothberg, the two define virtual reality as "any online digital platform—*Second Life*, of course, but also Instagram—in which the user

5. Ernest Cline, *Ready Player One*, 56.

6. "History of Virtual Reality," Virtual Reality Society, 2015, https://www.vrs.org.uk/virtual-reality/history.html.

7. "History of Virtual Reality."

8. Wagner James Au, *The Making of Second Life Notes from the New World* (Collins Business, 2009), 11.

9. "Oculus Rift: Step into the Game," Kickstarter, 2016, https://www.kickstarter.com/projects/1523379957/oculus-rift-step-into-the-game; "Google Inc., Google Cardboard," The Museum of Modern Art, accessed July 19, 2022. https://www.moma.org/collection/works/192718.

participates in a virtual world, as well as to the new media technology that bears this name."[10] While contemporary conversations about the metaverse often revolve around the sale of new products, to focus on one particular piece of hardware or software would miss the network for the nodes. When talking about the metaverse, we should look past the existing or forthcoming tech and instead consider our own stake in the future's development. What do *we*, as netizens of the evolving virtual world, want it to be?

In many ways, virtual reality—and the metaverse by association—pits the future defined by its citizens against a future defined by for-profit corporations. "To a certain extent, the tech industry writ large depends on futurism," writes Eric Ravenscraft. "Selling a phone is fine, but selling *the future* is more profitable."[11] This begs the question of whether we shape the future by creating the tools and technologies necessary to achieve our goals, or if the technologies created and imposed on us by tech giants like Meta are building the future on our behalf.

A more instructive guidebook might be science fiction that proposes possibilities for alternative societal systems. While novels of "hard" science fiction like *Ready Player One* focus on the future shaped by hardware, "soft" science fiction illuminates how new psycho-social frameworks might shape our collective experience. While it's clear why tech's billionaire class is drawn to hard sci-fi as fodder for future-building, soft sci-fi by authors like Octavia Butler, Ursula K. Le Guin, and Mark Dery show what kinds of worlds might be possible when we prioritize social questioning over technological evolution.

An alternative timeline of VR's development reveals the desire to understand how we see socially. In the 19th century, panoramic paintings and 360-degree murals allowed people to be immersed in an artificial field of vision. In 1933, the first published use of the term "virtual reality" is found in Antonin Artaud's *The Theater and its Double*. For him, "it refers rather to a particular potential for metamorphoses in states of matter and being that may be accessed via…theater."[12] In 1958, Charles "Chuck" Harrison—one of the few prominent Black designers of the time—popularized the View Master, a stereoscope with a manually-turning film reel on a cardboard disk.[13]

In 2006, Jaron Lanier and his team coined the term "homuncular flexibility" to describe how the human body gestures could guide the motion of unaccustomed avatar body forms.[14] As early as 2007, Nonny de la Peña, "the Godmother of VR," created the genre of "immersive journalism" and suggested the potential of the HMD as a device for empathy. In 2009, Skawennati, who is widely considered the first Indigenous net artist, created the machinima *TimeTraveller* about a young Mohawk man who uses time-traveling sunglasses from the 22nd century.[15] Radhika Gajjala's 2011 essay "Snapshots from Sari Trails: Cyborgs Old and New" explored secondary marketplace communities in *Second Life*.[16] In 2014, Ela Darling became the world's first VR porn performer.[17]

As a society, what might happen if we focused on the evolution of VR through the lens of its social implications? As Deborah Levitt writes in "Five Theses on Virtual Reality," "VR builds theories of

10. Deborah Levitt, "Five Theses on Virtual Reality and Sociality," *Public Seminar*, 2018, https://publicseminar.org/2018/05/five-theses-on-virtual-reality-and-sociality/.

11. Eric Ravenscraft, "What Is the Metaverse, Exactly?" *Wired*, 2022, https://www.wired.com/story/what-is-the-metaverse/.

12. Deborah Levitt, "Five Theses on Virtual Reality and Sociality."

13. "Charles Harrison, View-Master (Model G), 1962," The Museum of Modern Art, accessed July 19, 2022, https://www.moma.org/collection/works/205547.

14. Andrea Stevenson Won, Jeremy Bailenson, Jimmy Lee, and Jaron Lanier, "Homuncular Flexibility in Virtual Reality," *Journal of Computer-Mediated Communication* 20, no.3 (2015): 241–59, https://doi.org/10.1111/jcc4.12107.

15. Skawennati, "TimeTraveller™," timetravellertm.com, accessed July 19, 2022, https://www.timetravellertm.com/.

16. Radhika Gajjala, "Snapshots from Sari Trails: Cyborgs Old and New," *Social Identities* 17, no.3(2011): 393–408, https://doi.org/10.1080/13504630.2011.570977.

17. Jamie F, "The Evolution of VR Porn: Where Adult vr Started & Where It Is." (2021), Sex Tech Guide, May 28, 2021, https://sextechguide.com/vr/evolution-vr-porn-history/.

worlds through our experience of them. It enables us to use aspects of our perceptual systems in ways that other media do not to investigate problematics and scenarios."[18]

By using *Ready Player One* as a frame, we can more clearly discern the social complications of VR. By the final quarter of the novel, it's revealed that one the protagonists, Aech (pronounced "H")—assumed to be a white cis male—is actually a Black woman named Helen Harris. She claims it was her mother's idea to conceal her race and gender, as with user anonymity and a carefully crafted avatar, she would have more agency. "The OASIS was the best thing that had ever happened to both women and people of color. From the very start, [Helen's mother] Marie had used a white male avatar to conduct all of her online business, because of the marked difference it made in how she was treated and the opportunities she was given."[19]

This type of freedom found in race swapping has been part of the lore of virtual environments since its creation. In Lisa Nakamura's 1995 essay "Race In/For Cyberspace: Identity Tourism and Racial Passing on the Internet," she describes online role-playing sites that offer the participant the ability to select their race, gender, and physical appearance, which she frames as "identity tourism." "Players who elect to describe themselves in racial terms…are often seen as engaging in a form of hostile performance," she writes, "The borders and frontiers of cyberspace which had previously seemed so amorphous take on a keen sharpness when the enunciation of racial otherness is put into play as performance." The race-swapping participant, then, plays in a familiar, tokenized performance, appropriating alternate racial identities without the risks associated in physical life. In this way, it actually has the opposite impact of the "empathy" VR rhetoric; in fact, it reinforces stereotypes and a sense of otherness.

As the physical world becomes increasingly unlivable in *Ready Player One*, labor moves to the metaverse. Similarly, and thanks in large part to the work-from-anywhere tech culture prompted by the COVID-19 pandemic, Meta employees can now hold meetings in Horizon Workrooms—a virtual conference room with a wall of windows overlooking a waterfall. In Ben Tarnoff's 2021 essay "The Metaverse is a Cubicle," he suggests two primary reasons why, even with all the potential for virtual environments, employees might find themselves floating legless in a meeting room with an artificial view. The first reason for recreating the office environment is control. "Employers want to find a way to exercise managerial control over remote workers, to bring the collaborative and socially cohesive aspects of in-person work to remote/hybrid environments, and to push work to lower-wage regions."

The second is the illusion of freedom. "Employees want to retain flexibility around WFH while also finding a way to mitigate the unpleasant elements of WFH (Zoom fatigue, hybrid headaches, anxieties about lack of visibility leading to being passed over for promotions, etc.)." This virtual, metaphorical cubicle feels like the physical cubicle we've come to associate with the white-collar world, first introduced to give employees the feeling of having a personalized workspace without sparing them from round-the-clock managerial surveillance.

Critics claim that the metaverse is an old dream that will never achieve the ubiquity that its evangelists predict. "VR is a bit like a rich white kid with famous parents: It never stops failing upward, forever graded on a generous curve, always judged based on its 'potential' rather than its results," writes David Karpf in *Wired*.[20] There are a couple reasons that this time, in 2022, might be different. Notably, the technology has improved dramatically. Even if avatars still do not have legs in Horizon, the hardware is light and fast, and likely to advance with

18. Deborah Levitt, "Five Theses on Virtual Reality and Sociality."

19. Ernest Cline, *Ready Player One*, 320.

20. David Karpf, "Virtual Reality Is the Rich White Kid of Technology," *Wired*, 2021, https://www.wired.com/story/virtual-reality-rich-white-kid-of-technology/.

over $18.5 billion a year spent on VR/AR R&D by Meta alone. And, as Tarnoff notes, with remote and hybrid structures finding permanence as the pandemic wanes, people (both employers and employees) are seeking new ways of connecting and working online.

Assuming the metaverse will only grow in prominence moving forward, now is when we should begin to shift our focus from the "hard" technologies of VR towards the "soft" structures that can be re-envisioned to create a better future. "VR helps to shift our powerful cultural attachments to particular notions of individual and social group boundaries through experiences that, while never the experiences of particular others as the empathy machine logic would have it, reveal the flexibility of the human itself in relation to (its) others."[21] This shift returns to the origins of the phrase "virtual reality" itself, as described by Artaud in his proposition of theater. Theater and storytelling allowed for metamorphosis, a new potential for representation. We don't need new technologies, describes James Bridle, but "new metaphors: a metalanguage for describing the world that complex systems have wrought."[22] The fictions we share ultimately shape future pathways. The focus, therefore, is not on the new or different, but on how these lead to the potential for change.

In the final pages of *Ready Player One*, Halliday laments, "[A]s terrifying and painful as reality can be, it's also the only place where you can find true happiness. Because reality is *real*."[23] But reality and virtual reality is not an either/or binary. Virtual experiences are informed and shaped by physical experiences. As our digital technologies continue to enhance, driven by nostalgia and platform capitalism alike, what we need are new perceptual structures and soft approaches to worlding.

21. Deborah Levitt, "Five Theses on Virtual Reality and Sociality."

22. James Bridle, *New Dark Age: Technology and the End of the Future* (Orca Book Services, 2019).

23. Ernest Cline, *Ready Player One*, 364.

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